8. Secular Stagnation

Adv. Macro: Heterogeneous Agent Models

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Introduction

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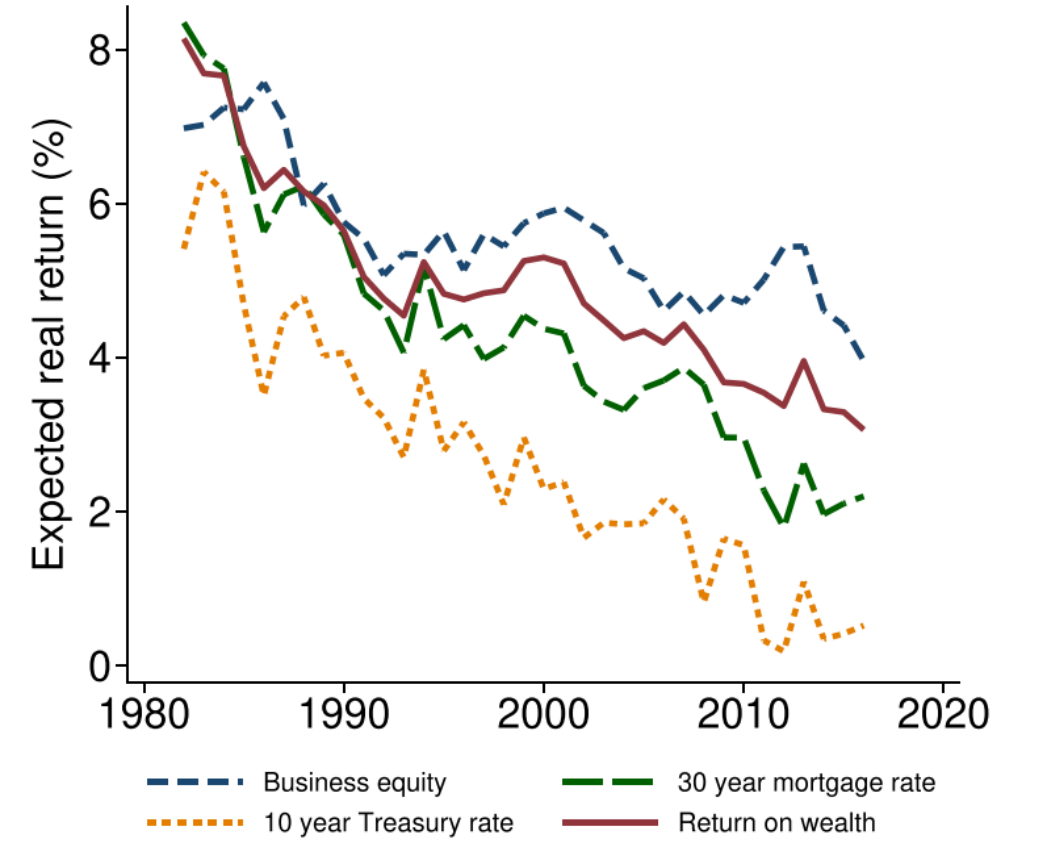
- **Plan for today:**
  1. Income inequality and saving (Straub 2019)
  2. The race between asset supply and asset demand (Auclert, Malmberg, Rognlie and Straub 2025)
  3. Indebted demand (Mian, Straub and Sufi 2021)

Secular Stagnation

- **Secular stagnation**: A state in which private demand is structurally low
  - Low level of growth in the economy
  - Low interest rates
  - Low level of inflation
- Large literature suggests that advanced economies have been in this state over the past $\approx$ 20 years

Declining interest rates

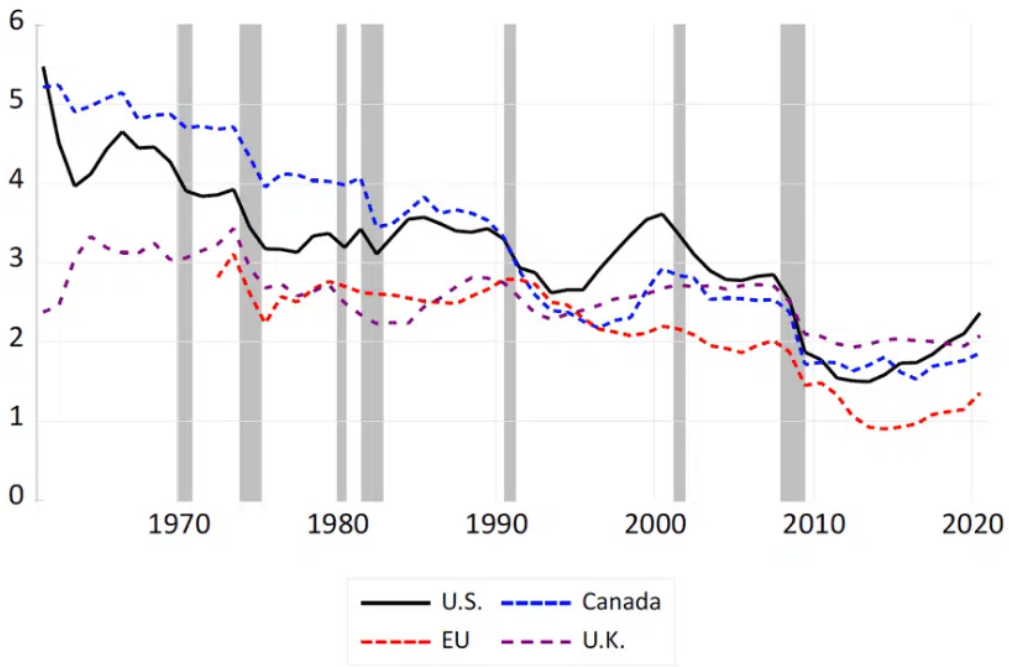
Various interest rates from Mian, Straub and Sufi (2021)



Declining growth

Trend Real Output Growth, 1961 to 2020

Percent



Source: www.cbo.gov/system/files/2020-12/56891-real%20-interest-rates.pdf

Data sources: Holston et al. (2017) and Federal Reserve Bank of New York, "Measuring the Natural Rate of Interest" (accessed December 10, 2020), www.newyorkfed.org/research/policy/rstar.

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Implications for fiscal and monetary policy? Low r means more likely to hit ZLB and fiscal stimulus cheap and powerful.

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- All of these in one framework: Auclert, Malmberg, Rognlie & Straub (2025)

Inequality and saving
Straub (2019)

Secular stagnation and income inequality

- A literature explains the decline in r through *rising* income/wealth **inequality**
  - See e.g. Mian, Straub and Sufi (2021): *What explains the decline in r*? Rising income inequality versus demographic shifts*

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- **Example**: If saving rates increase with income (richer save more) then:
  - Redistribution from poor to rich households *increase* the aggregate supply of savings $\Rightarrow$ lower rates in GE
  - Suggests link between increasing income inequality and secular stagnation
  - Straub (2019, WP) tests this hypothesis

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$$\max_{c_t, a_{t+1}} u(c_t) + \beta U(a_{t+1})$$

$$c_t + R^{-1} a_{t+1} \leq a_t + w_t$$

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We can show that $c_t \approx k(a_t + w)^{\phi}$, with $\phi = \frac{\Sigma}{\sigma}$

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- Two questions
  - What does this elasticity look like empirically?
  - If $\phi \neq 1$ how can we accommodate this in a HA model?

Empirical estimate

- Use US panel survey data (PSID)

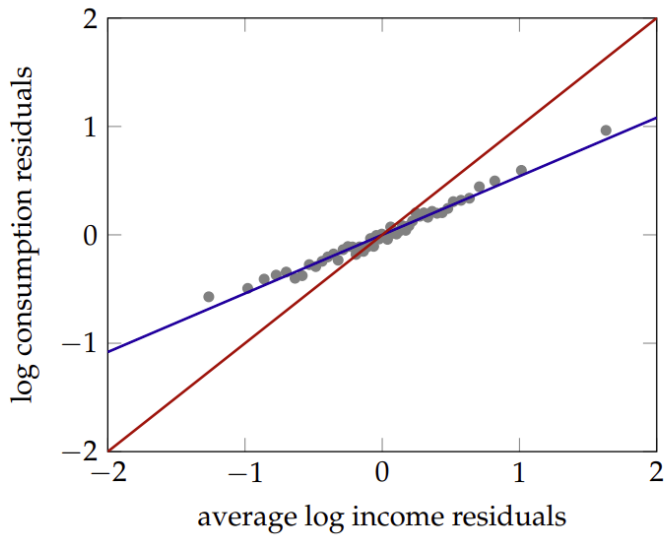
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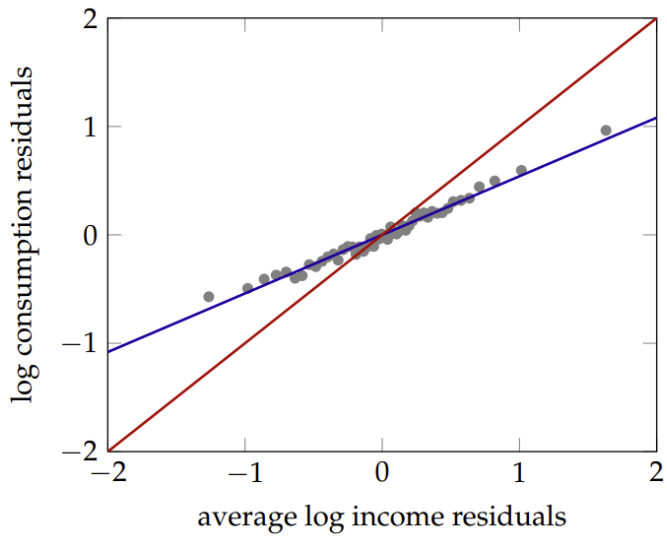
Figure 3: Consumption and average income.



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- More detailed estimates in paper, suggest $\phi \approx 0.7$

Permanent redistribution in the canonical HA model

- Consider standard HA model with permanent income state p :

$$V(a_{t-1}, z_t, p) = \max_{c_t} \frac{c_t^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}\left[V(a_t, z_{t+1}, p)\right]$$

$$\text{subject to}$$

$$c_t + a_t = R a_{t-1} + z_t p$$

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- Note: preferences are *homothetic* (homogeneous of degree 1)
  - »Scale independent«

Homothetic household problem I

- Normalize constraints by p and Bellman by $p^{1-\sigma}$

$$\frac{V\left(a_{t-1}, z_t, p\right)}{p^{1-\sigma}} = \max_{c_t} \frac{\frac{c_t^{1-\sigma}}{1-\sigma}}{p^{1-\sigma}} + \beta \frac{\mathbb{E}\left[V\left(a_t, z_{t+1}, p\right)\right]}{p^{1-\sigma}}$$

subject to

$$\frac{c_t}{p} + \frac{a_t}{p} = R\frac{a_{t-1}}{p} + z_t\frac{p}{p}$$

$$\frac{a_t}{p} \geq \frac{0}{p}$$

Homothetic household problem II

Define $\tilde{c}_t = \frac{c_t}{p}, \tilde{a}_t = \frac{a_t}{p}, \tilde{V}_t = \frac{V_t}{p^{1-\sigma}}$ to get:

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- Permanent redistribution has *no effect* on aggregates: the lower saving of one group exactly offsets the higher saving of the other

Non-homothetic HA model

- Extend standard HA model with *taste for wealth* (»status«)

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- If $\sigma \neq \sigma_a$, the weight $p^{\sigma-\sigma_a}$ on wealth depends on p: richer households put more weight on wealth when $\sigma > \sigma_a$

Applications

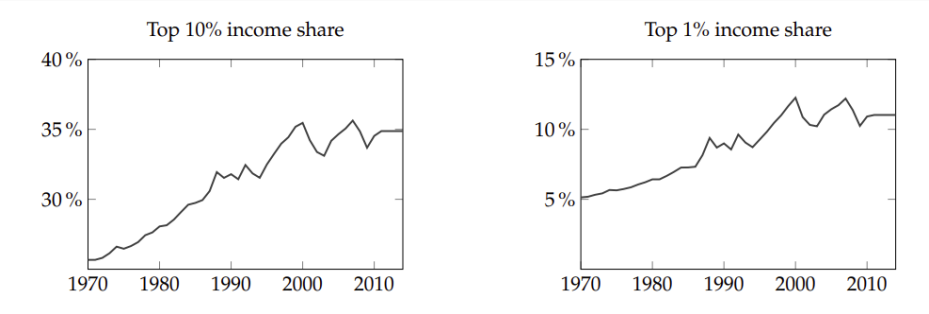
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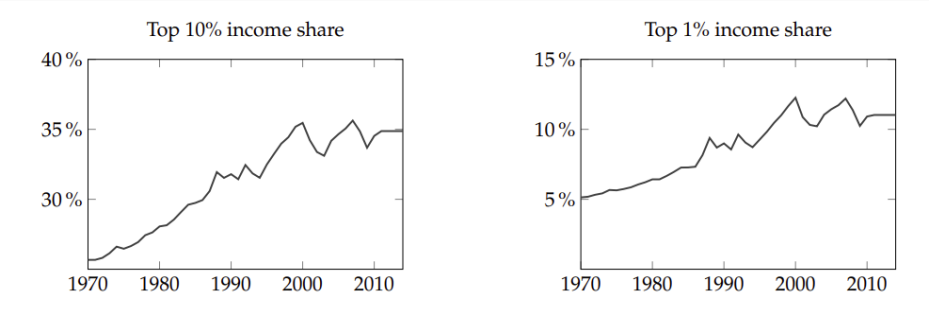
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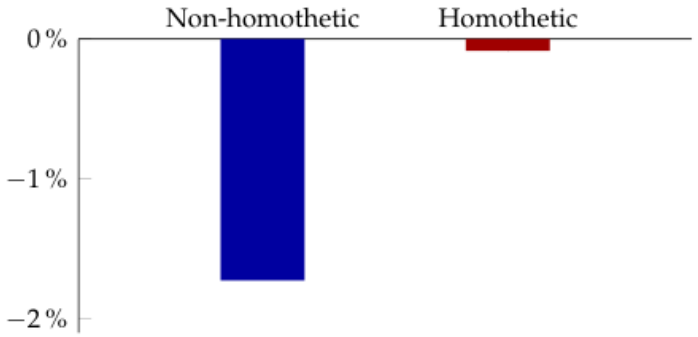


- Solve for GE (Supply side as in HANC)

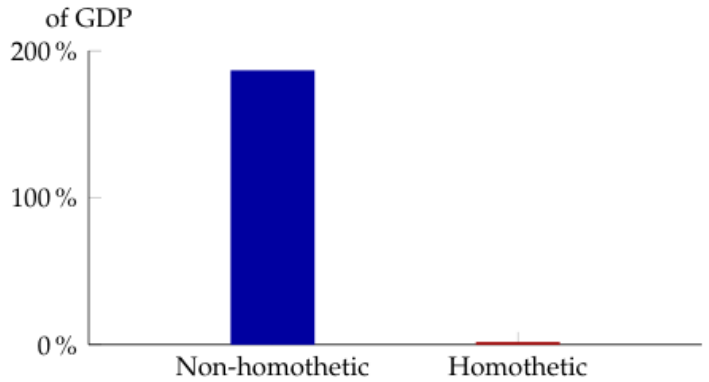
Results

Figure 6: The partial equilibrium effects of a rising income inequality.

(a) Short-run drop in aggregate consumption.

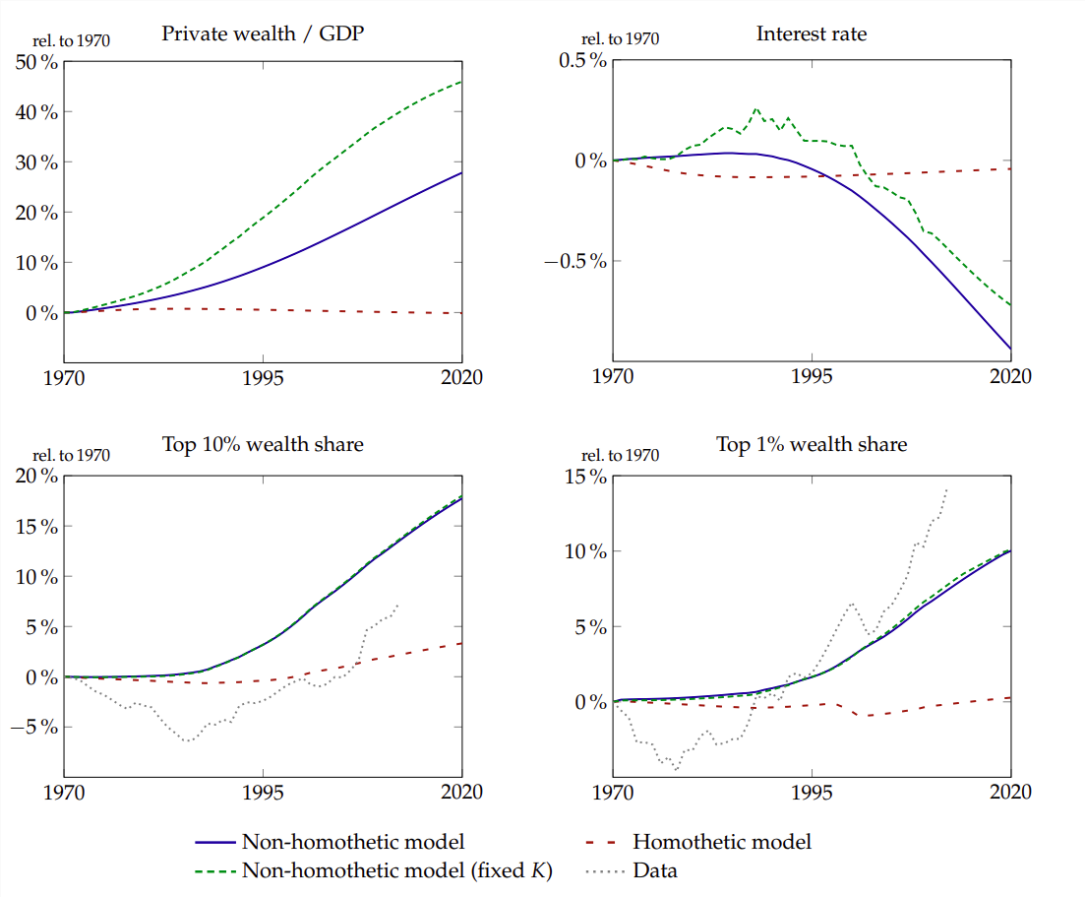


(b) Long-run rise in aggregate savings.



GE implications of rising income inequality

- Compare effects in homothetic and non-homothetic models:



- Note: »Fixed K« is Lucas-tree economy, where wealth grows due to rising asset prices

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- Increase in savings might not finance more physical capital, but just higher prices of financial assets!

Asset supply and demand

Auclert, Malmberg, Rognlie and Straub (2025)

The race between asset supply and asset demand

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  - fertility fell from over 3 children per woman to 1.6: the population ages
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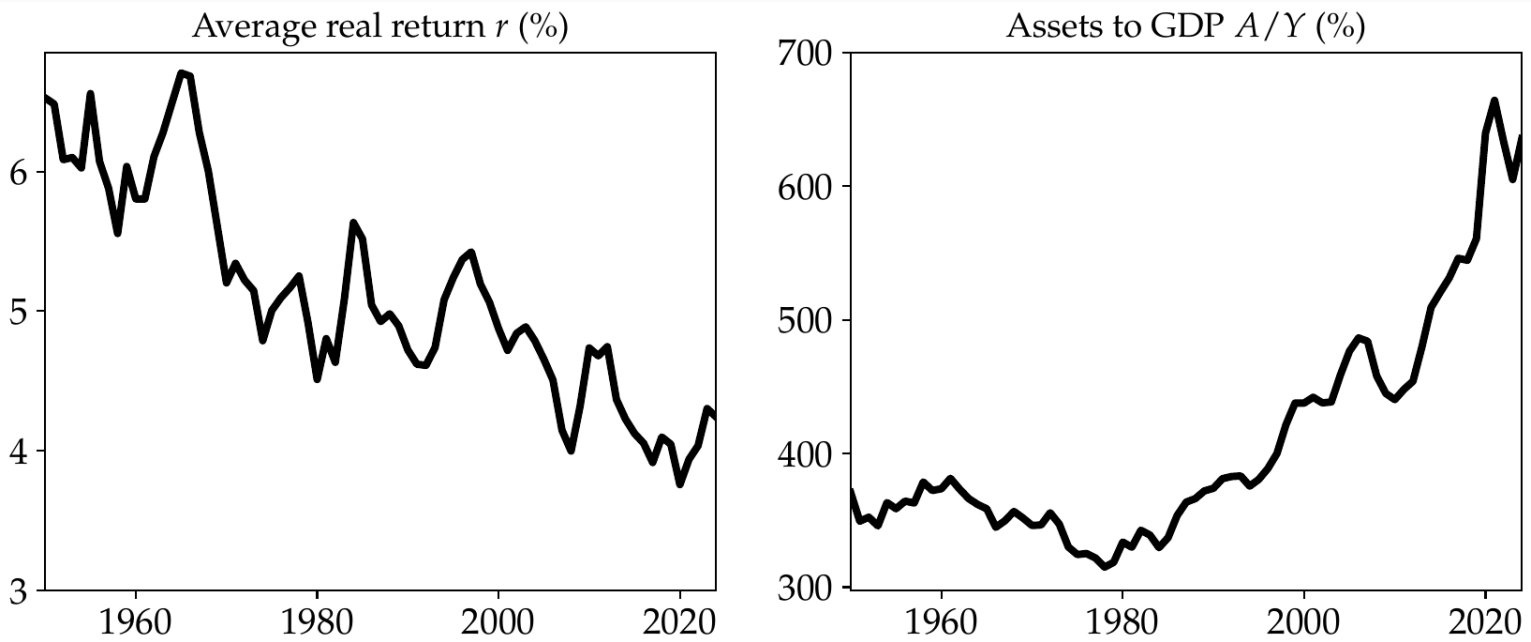
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  1. How did these shifts affect interest rates and wealth over the last 75 years?
  2. What happens over the next 75 years, and what does it mean for public debt?
- Method: long-run asset supply and demand, measured with *sufficient statistics* from a life-cycle model
  - extends Auclert, Malmberg, Martenet and Rognlie (2024), who study demographics only

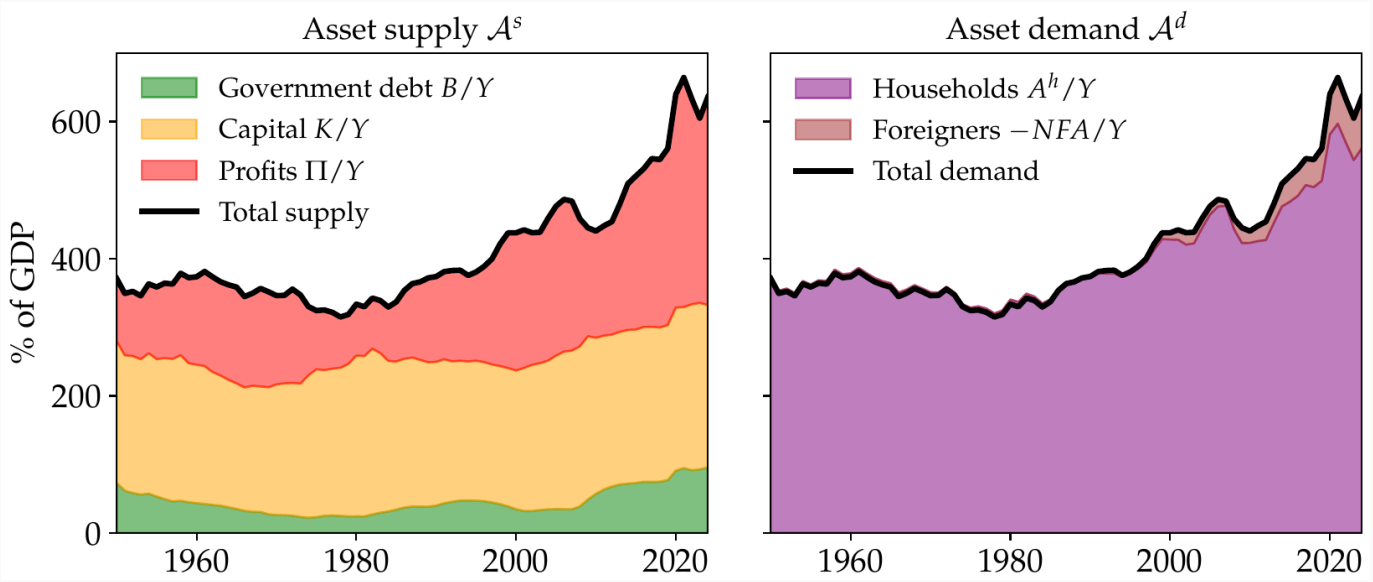
Falling returns and rising wealth



- Average return $r = \bar{s}r^f + (1 - \bar{s})r^r$: safe rate on government bonds r^f and return on private assets r^r, with $\bar{s} = 0.11$
- Since the 1950s, r fell by about 2 pp and wealth rose by about 300% of GDP, mostly after 1980

Source: AMRS (2025), Figure 2

Realized asset supply and demand, 1950-2024



- Supply: government debt B, capital K, capitalized profits and rents Π. Demand: households A^h and foreigners $-NFA$
- B, K, Y from the national accounts, A^h and NFA from the financial accounts, Π as the residual
- Supply rose through government debt and profits, demand through households and foreigners

Source: AMRS (2025), Figure 3

Asset market equilibrium

- The long-run rate r^* clears the market for assets, relative to GDP:

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- Vocabulary: household saving is a *demand* for assets. Same diagram as Aiyagari (lectures 3-4), with B and Π added

Measuring the shifts

- Let η shift both curves, and define the semi-elasticities

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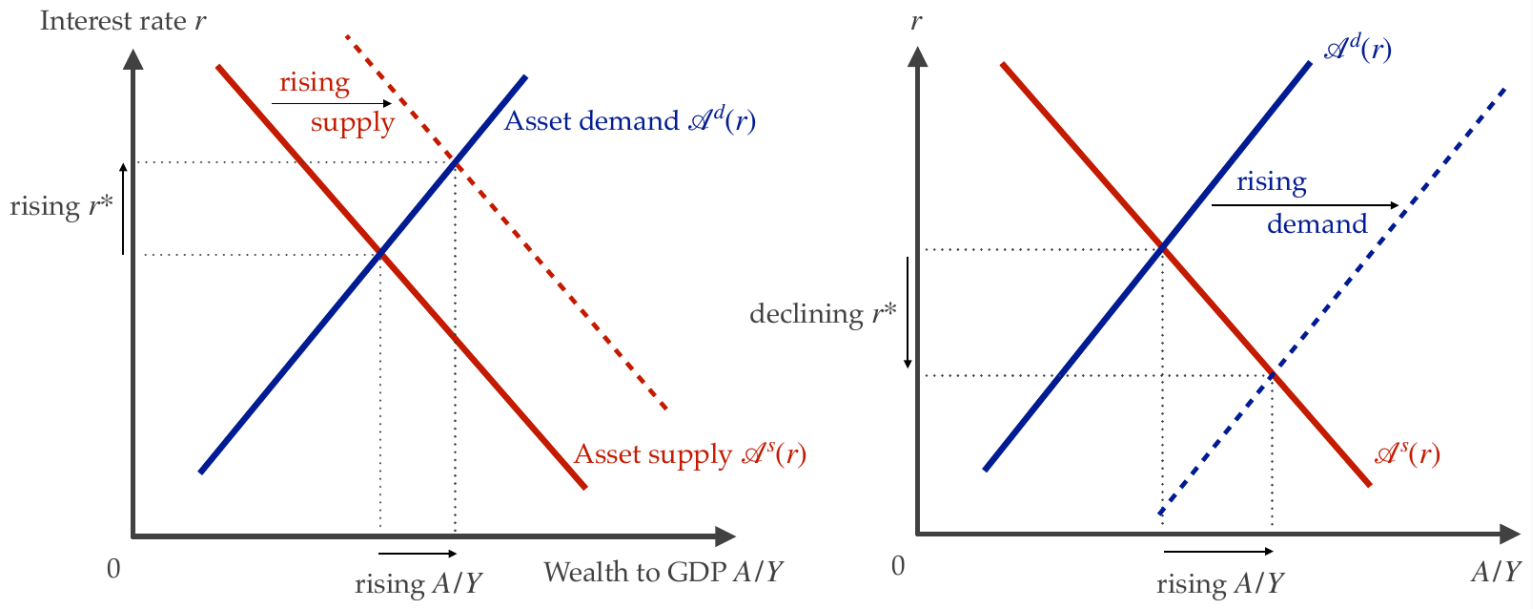
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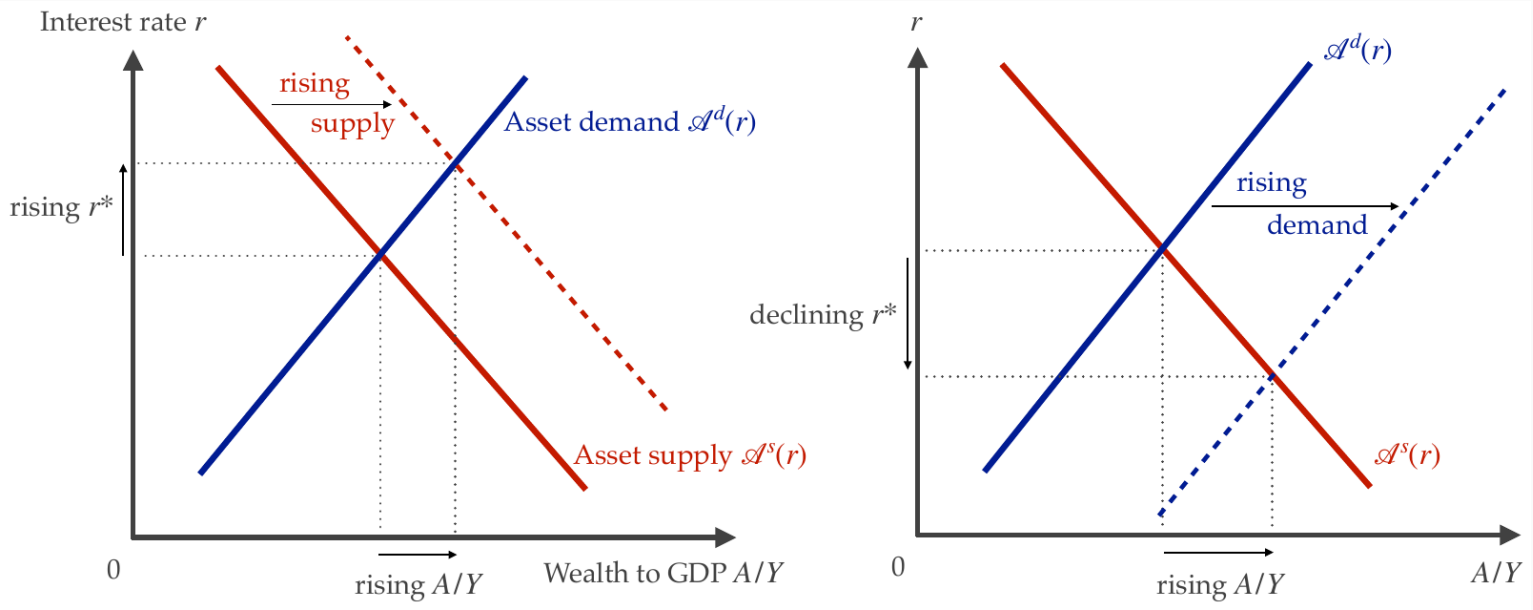
- Only two numbers needed: $\epsilon^s \approx 20$ and $\epsilon^d \approx 9$ (from the model below)

Who won the race?



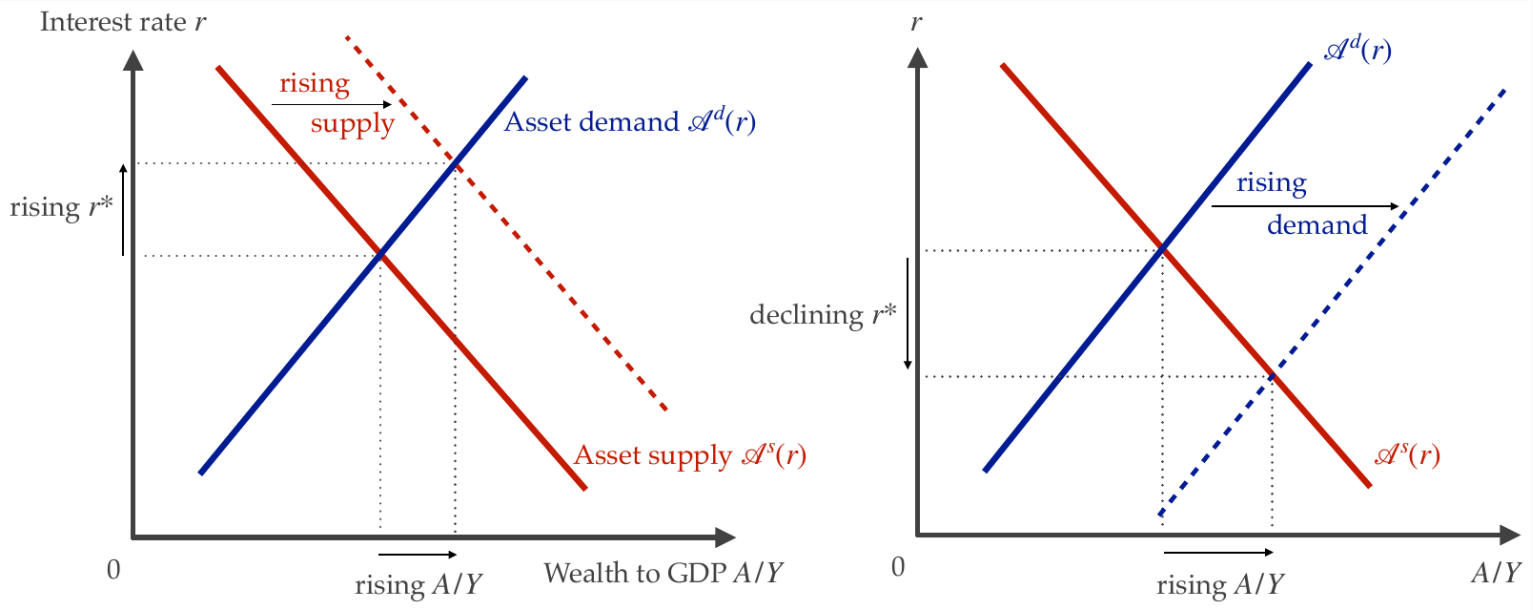
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- Data: A/Y up and r down $\Rightarrow$ asset demand won the race

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- Solve the firm's problem from the FOC: Cobb-Douglas, markup μ, tax τ on profits net of depreciation. Capital FOC: $\frac{\alpha}{\mu}\frac{Y}{K} = \frac{r^r}{1-\tau} + \delta$, profits are a share $1 - 1/\mu$ of output

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Model: households in the data

- **Wealth by age and type** is read from the data, not computed:
  - by age: 2016 profiles of net worth a_j and labor income we_j (SCF)
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- **Interpretation**:
  - $\mathbb{E}Age_a$: age of the owner of the average dollar of wealth
  - $\mathbb{E}Age_{\tilde{c}}$: age at which the average dollar is consumed
  - $\mathrm{Var}Age_{\tilde{c}}$: spread of consumption over the life cycle, the room to substitute

The slope of asset demand

Proposition 2 (with $r = g$)

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- With $\sigma = 1/3$, $\epsilon^d \approx 9$. Representative agent: $\epsilon^d = \infty$, so asset demand shifters cannot move r^*

Model: shifters of asset demand

- **Compositional**: change the weights on ages and types, not behavior. Reweight the 2016 data profiles
  - + **Demographics**: an older population puts more weight on asset-rich, low-income ages
  - + **Productivity growth**: slower growth across cohorts makes older cohorts relatively richer, like aging
  - + **Inequality**: income moves to the rich, who hold more wealth relative to their income, as in Straub (2019)

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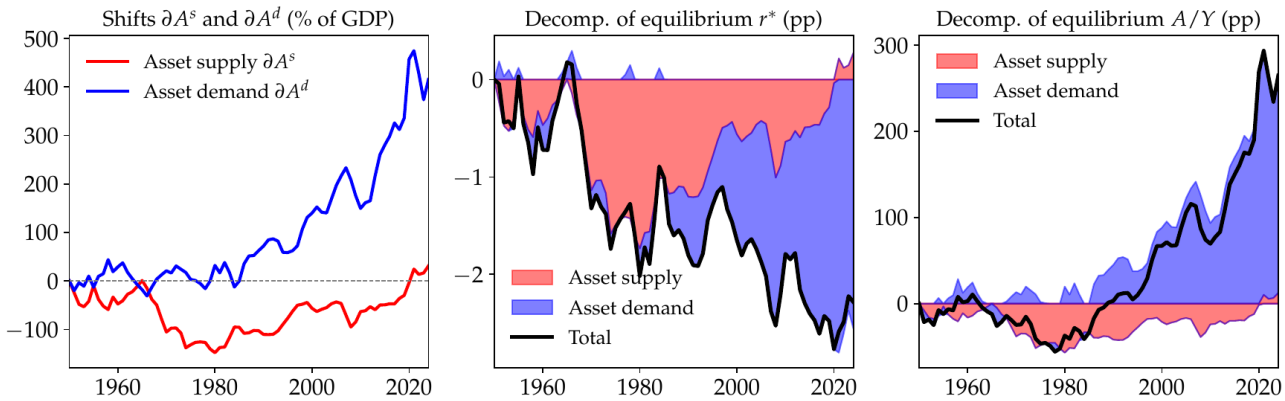
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Asset demand won the race



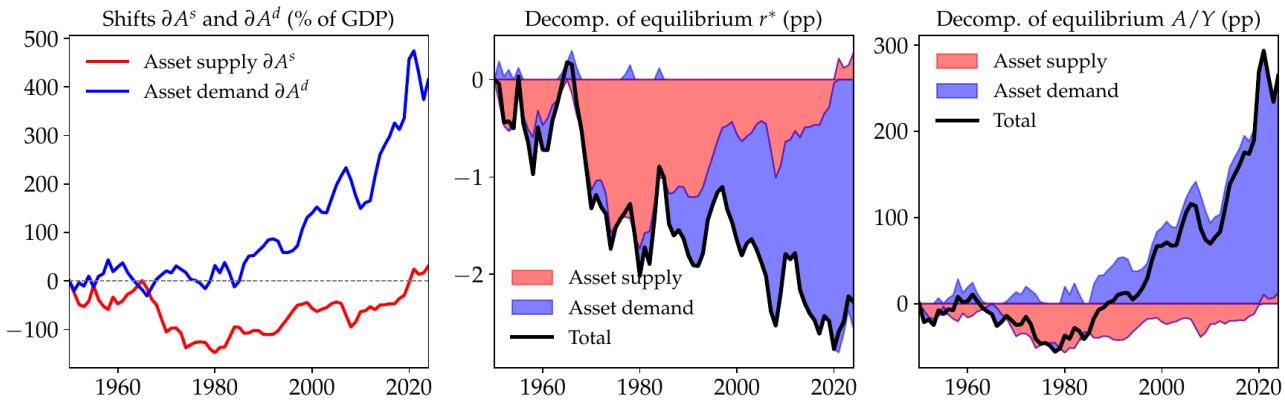
- Each year, two observations (dr, $d \log A/Y$ since 1950) and two unknown shifts:

$$\partial \log \mathcal{A}^s = d \log \frac{A}{Y} + \epsilon^s dr, \qquad \partial \log \mathcal{A}^d = d \log \frac{A}{Y} - \epsilon^d dr$$

change in A/Y = shift of the curve + move along it caused by dr

Source: AMRS (2025), Figure 4

Asset demand won the race



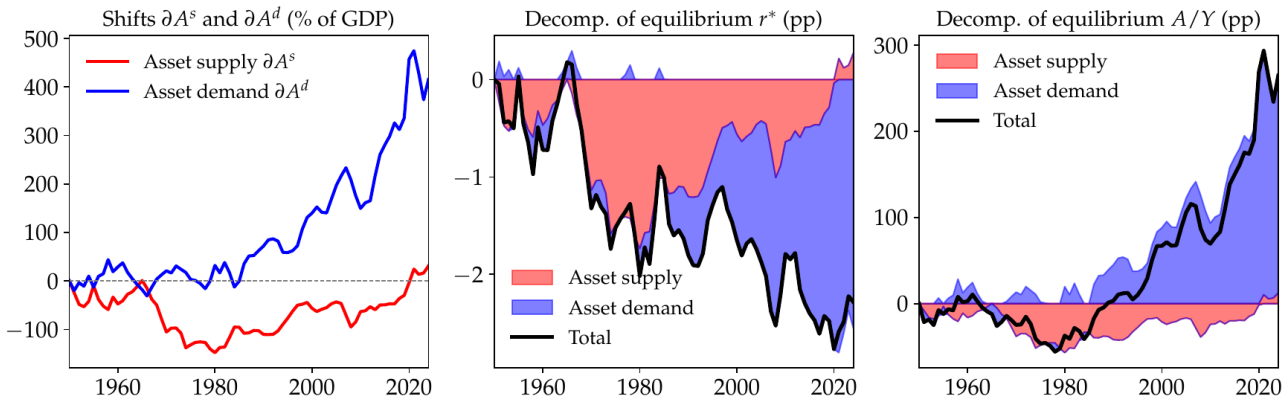
- Each year, two observations (dr, $d \log A/Y$ since 1950) and two unknown shifts:

$$\partial \log \mathcal{A}^s = d \log \frac{A}{Y} + \epsilon^s dr, \qquad \partial \log \mathcal{A}^d = d \log \frac{A}{Y} - \epsilon^d dr$$

  change in A/Y = shift of the curve + move along it caused by dr
- r fell by about 2 pp. Along supply, A/Y rises by $\epsilon^s \times 0.02 \approx 40$ log points, most of the observed rise $\Rightarrow$ small supply shift

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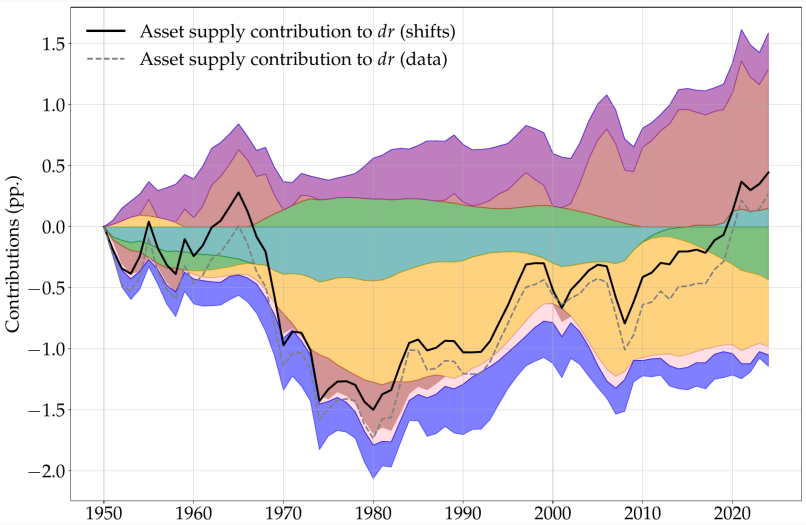
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- Along demand, A/Y falls by $\epsilon^d \times 0.02 \approx 18$ log points, yet A/Y rose $\Rightarrow$ large demand shift

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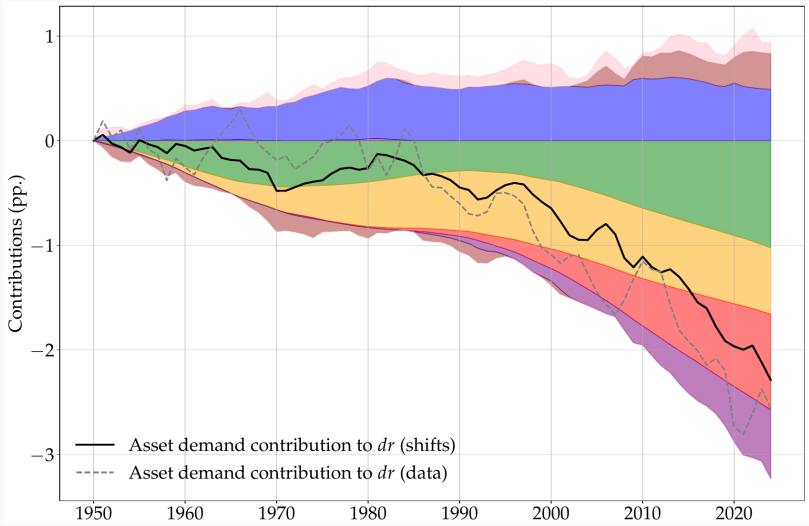
Which shifters moved r^*?

Asset supply



- Government debt
- Demographics
- Productivity growth
- Markups and rents
- Taxes
- Risk premium
- Capital intensity

Asset demand



- Demographics
- Productivity growth
- Inequality
- NFA
- Social security
- Labor share
- Taxes

Contribution of each shifter to $dr^* = \left(\partial \log \mathcal{A}^s - \partial \log \mathcal{A}^d\right) / (\epsilon^s + \epsilon^d)$, in pp.
Solid: sum of the shifters. Dashed: shift backed out from the data

Source: AMRS (2025), Figures 9(b) and 10(b)

The race behind us, 1950-2024

Asset supply

	% GDP	bp
Government debt	23	15
Demographics	−65	−43
Productivity growth	−82	−55
Markups and rents	170	114
Taxes	−10	−7
Risk premium	45	30
Capital intensity	−14	−9
Total explained	67	45
Total data	31	27

Asset demand

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Demographics	145	−103
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Labor share	−48	34
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Total explained	310	−229
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Shift of each curve (% of GDP) and contribution to dr^* (bp)

- Demographics and inequality: about −1 pp each. Slower growth and foreign inflows: about −0.7 pp each

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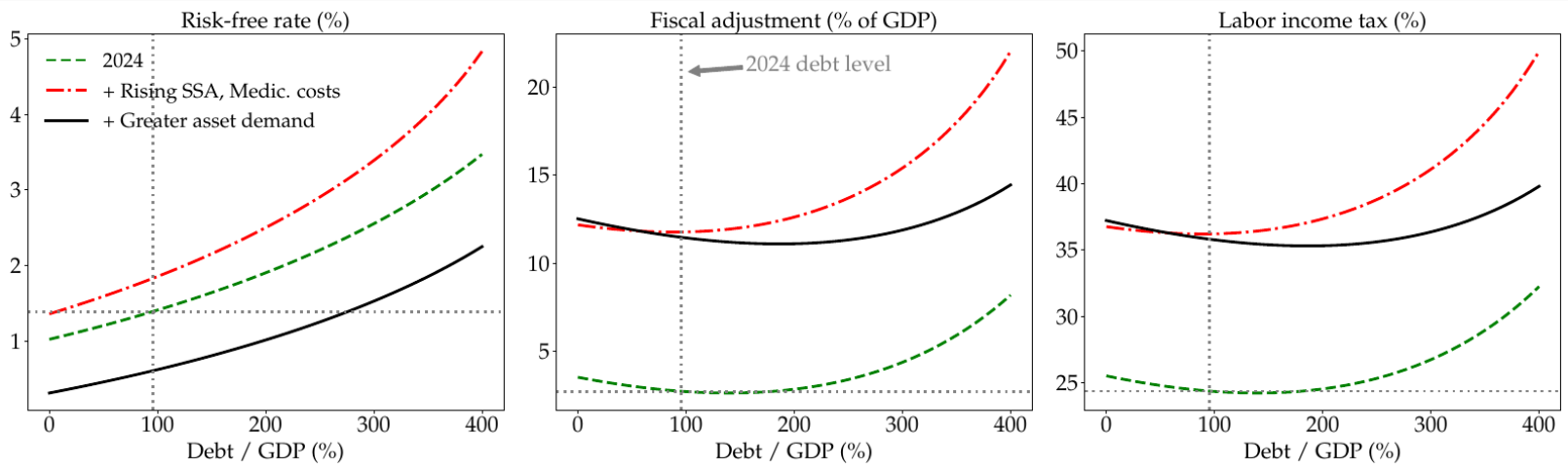
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Shift of each curve (% of GDP) and contribution to dr^* (bp)

- Demographics and inequality: about −1 pp each. Slower growth and foreign inflows: about −0.7 pp each
- Markups and rents held r^* up by more than 1 pp: capitalized profits are an asset that households must hold

Source: AMRS (2025), Tables 1 and 2

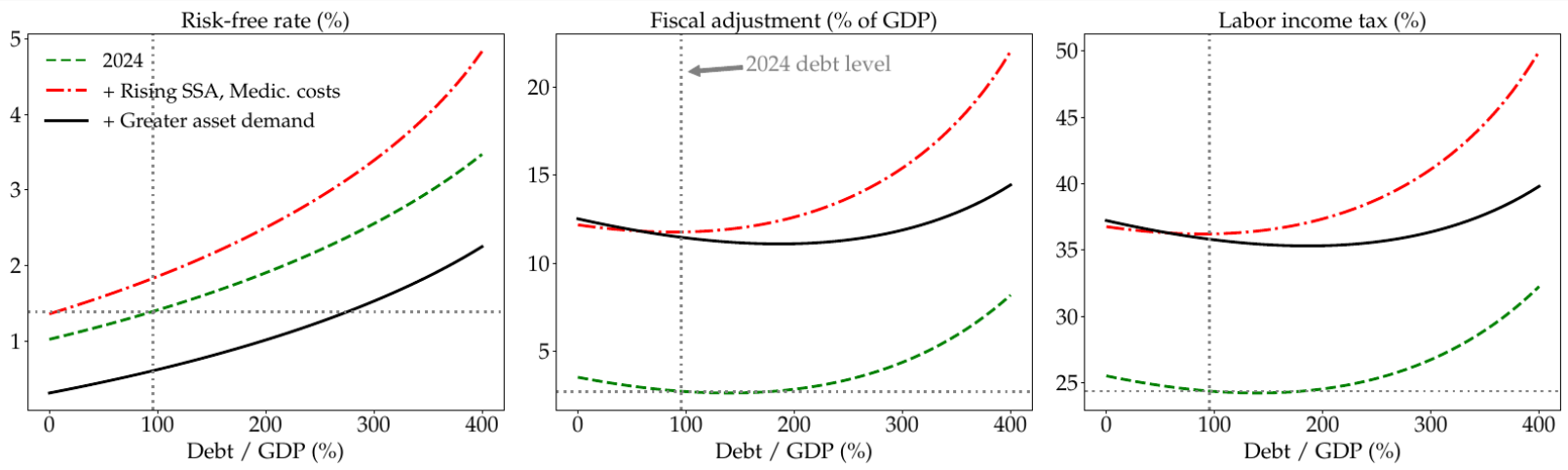
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- Aging keeps raising asset demand: with fixed supply, r^* falls by 1.2 pp by 2100

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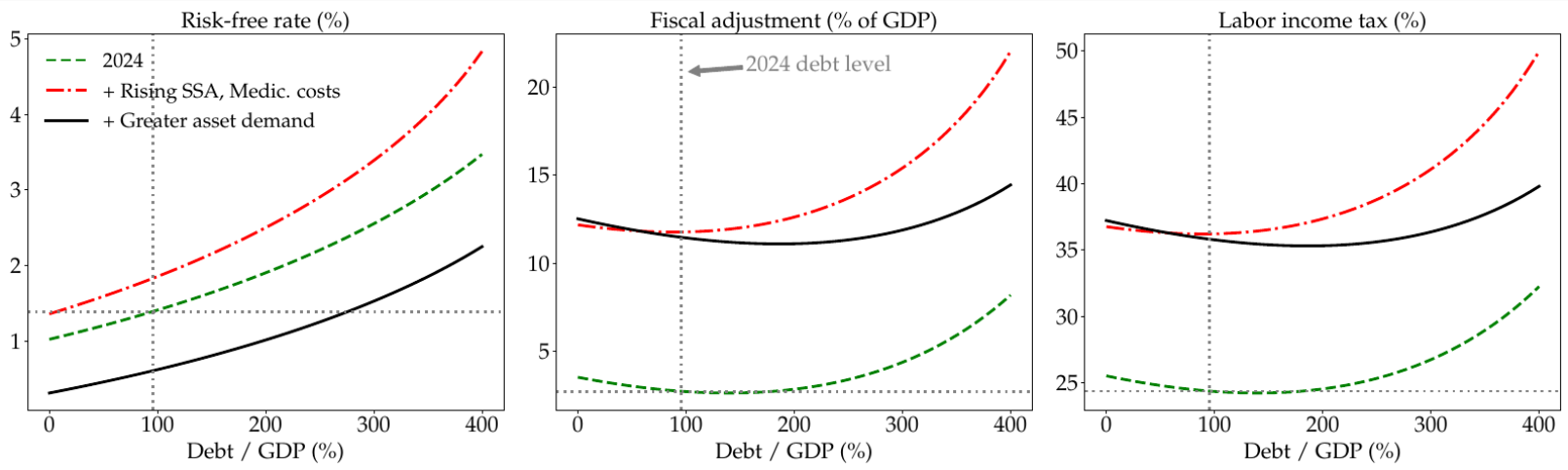
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The race ahead



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- But old-age spending rises from 11% to 20% of GDP: without adjustment, debt grows by 500% of GDP and supply wins
- Debt stabilized in 2100 (figure): 250% of GDP at today's r, but it needs an adjustment of about 12% of GDP

Source: AMRS (2025), Figure 15

Takeaways

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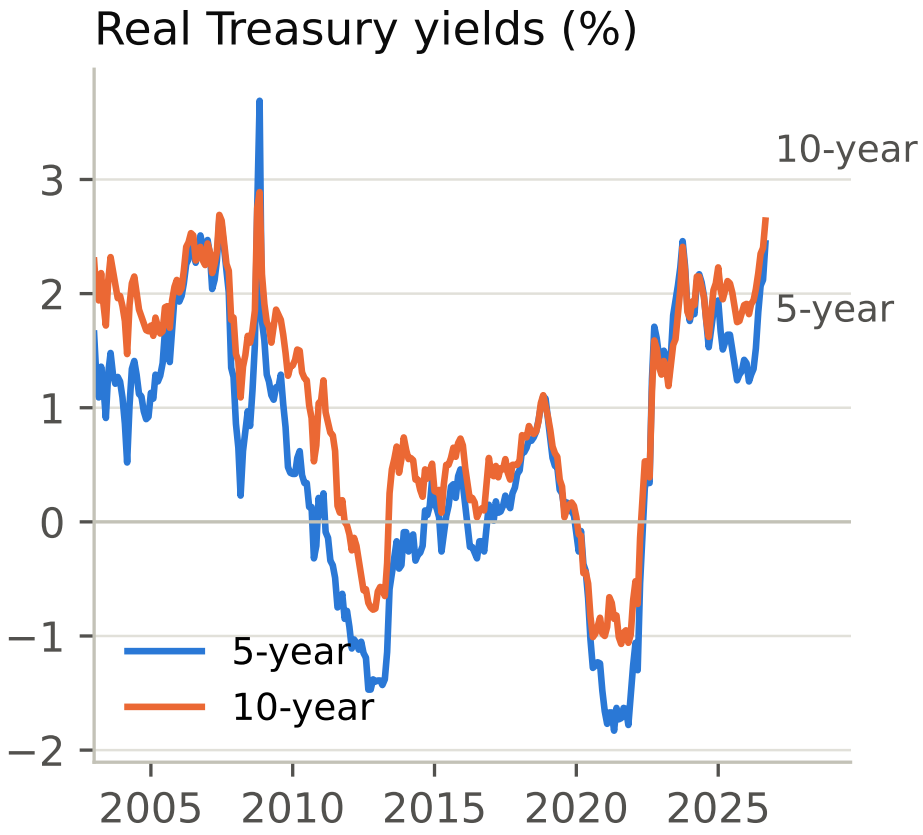
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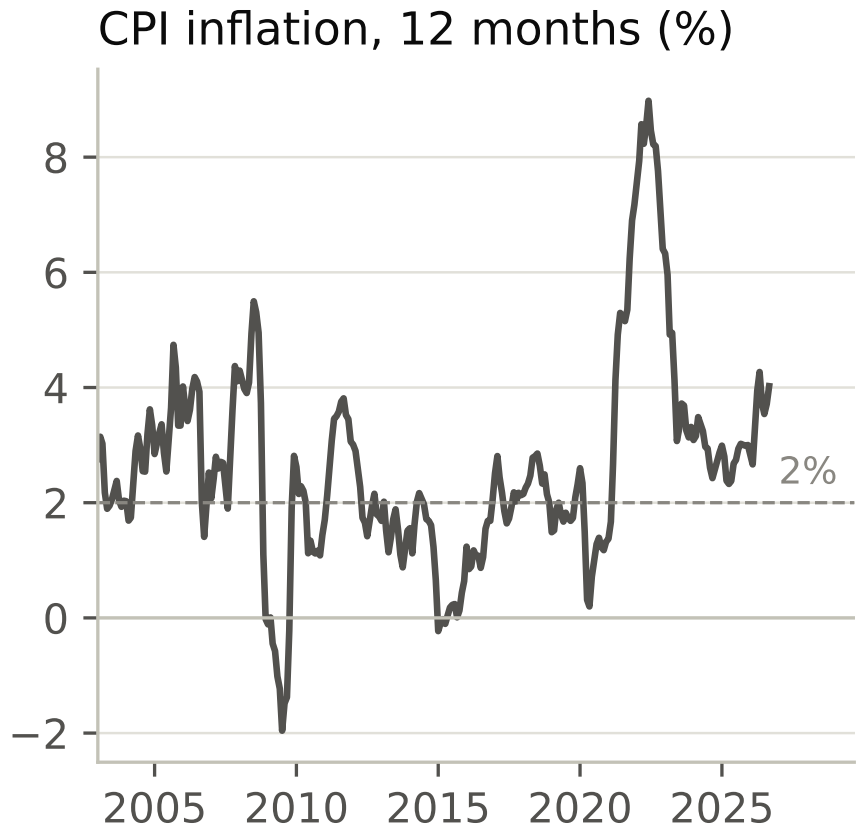
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- Heterogeneity makes asset demand imperfectly elastic, so aging and inequality move r^*
- 1950-2024: asset demand won by far (aging, inequality, slower growth, foreign inflows), and r^* fell
- Going forward: aging raises asset demand, but also deficits. There is room to borrow, but a large fiscal adjustment is unavoidable

Higher interest rates and inflation: a new normal?





Source: FRED (FII5, FII10, CPIAUCSL), monthly.

Indebted demand
Mian, Straub and Sufi (2021)

Introduction

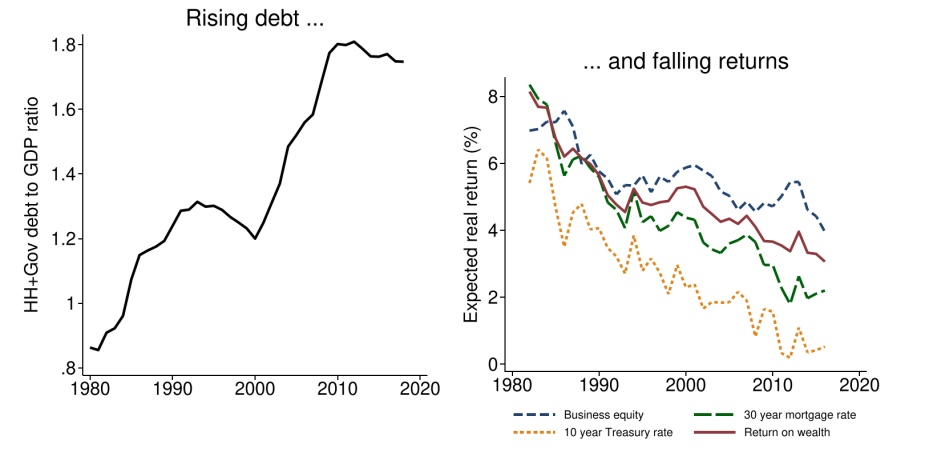
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- Highlights role of increasing household debt and connect to returns

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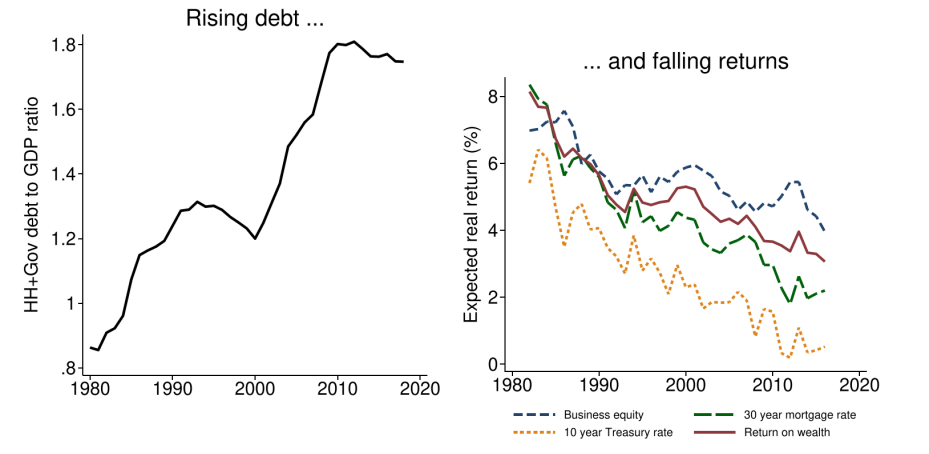
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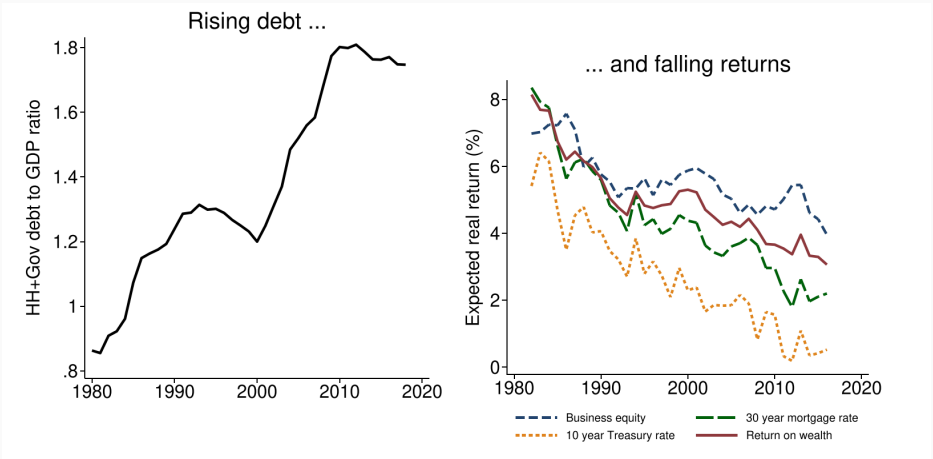
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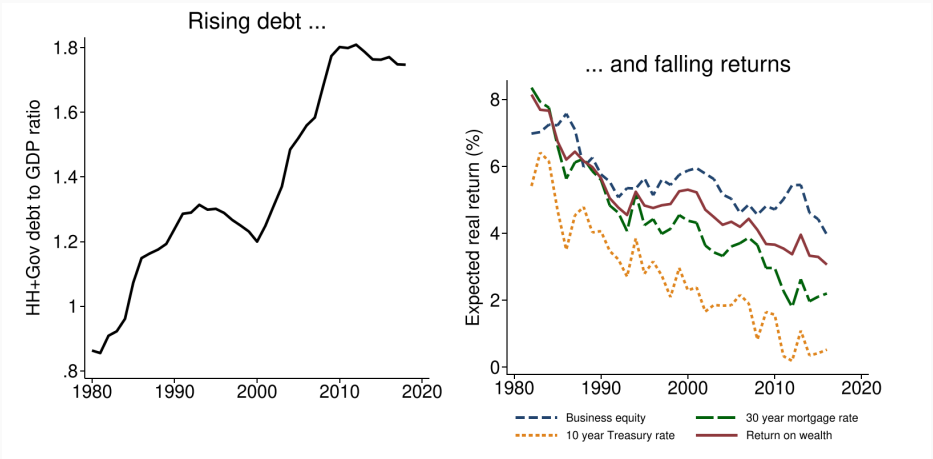
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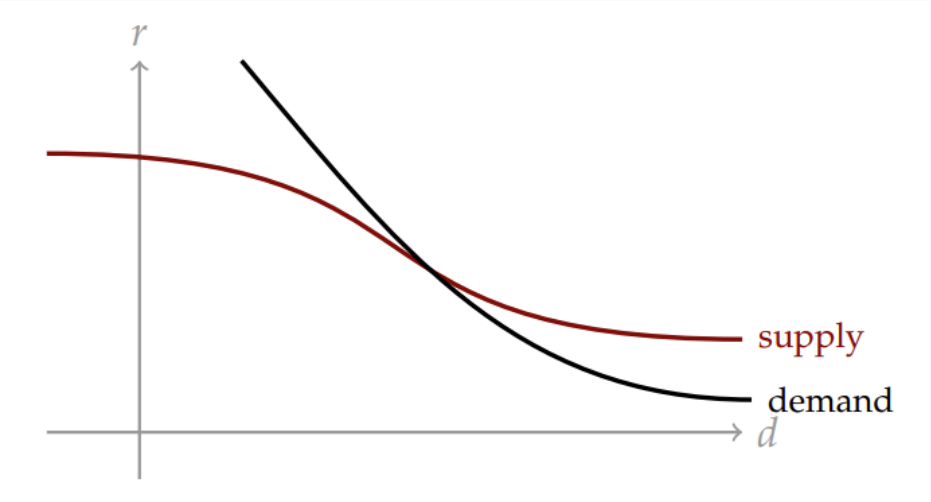
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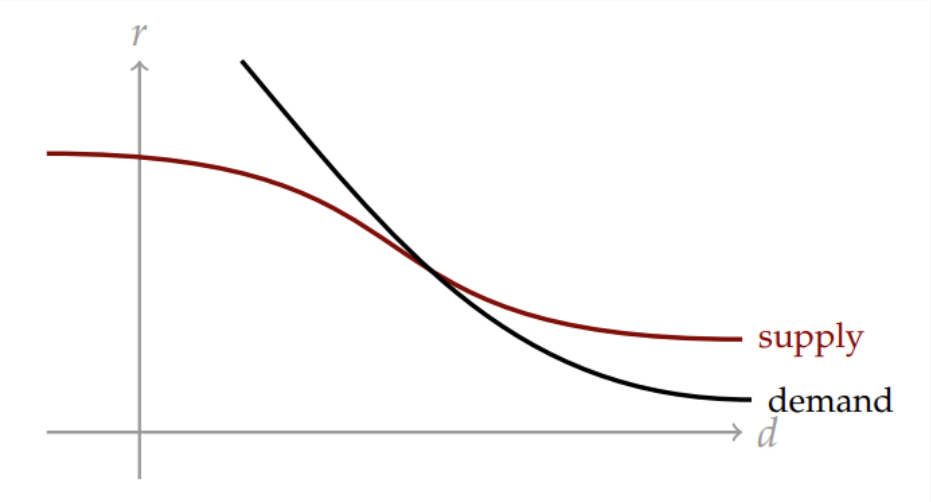
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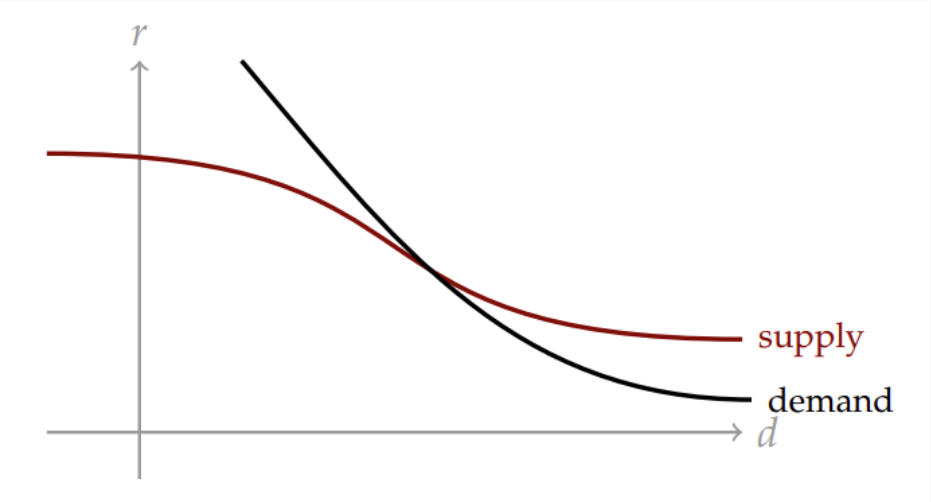
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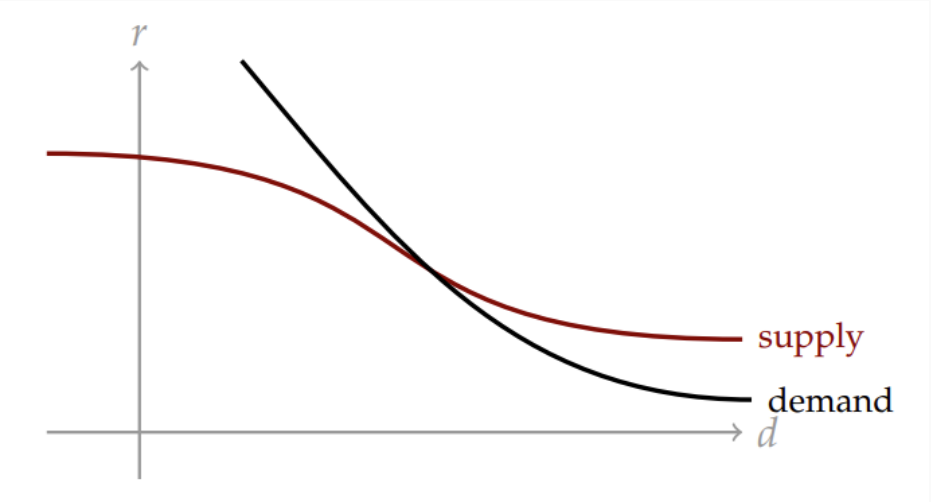
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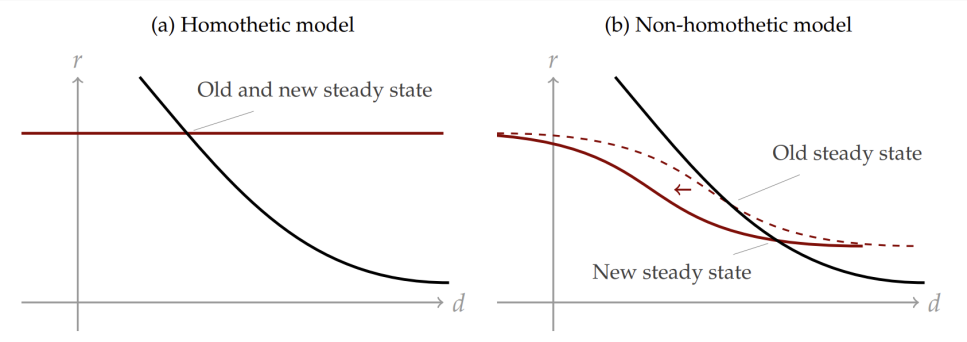
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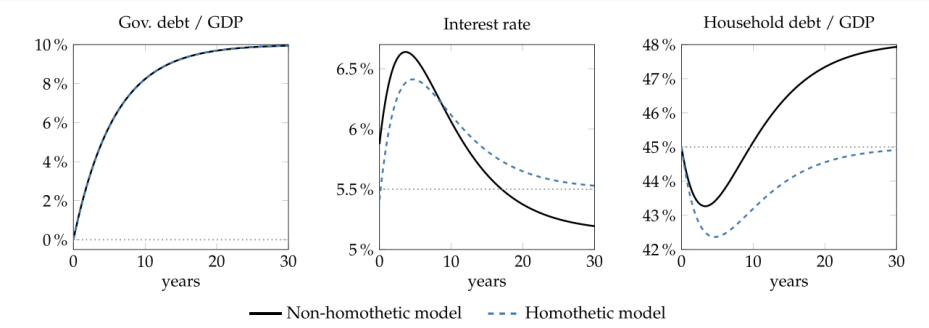
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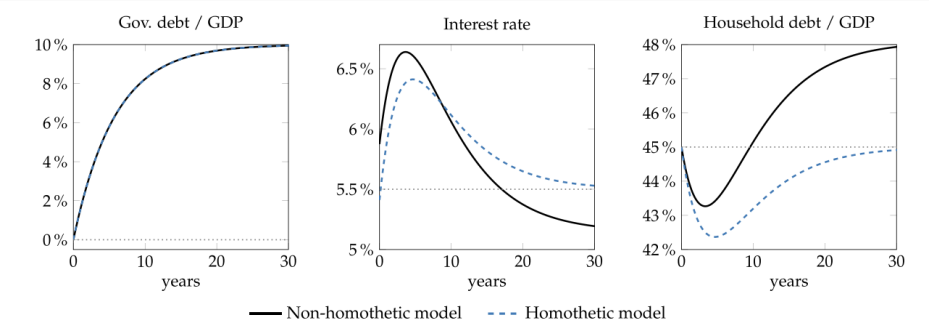
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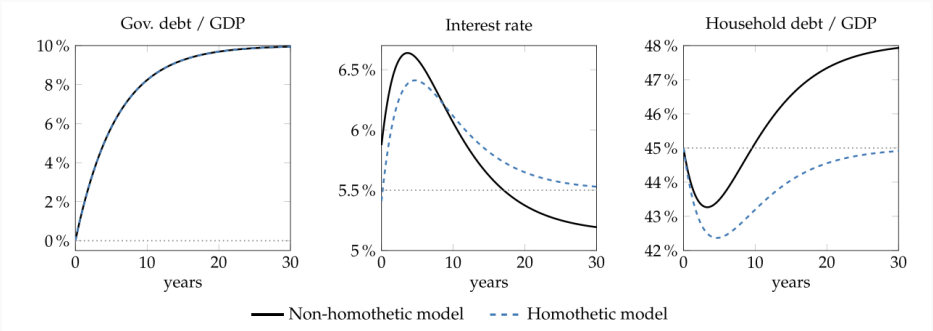
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- For monetary policy: lower r means ZLB more likely, "less ammunition"

Exercise

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- Consider the following PE HA model:

$$V(a_{t-1}, z_t, p) = \max_{c_t} \frac{c_t^{1-\sigma}}{1-\sigma} + \phi_a \frac{a_t^{1-\sigma_a}}{1-\sigma_a} + \beta \mathbb{E}\left[V(a_t, z_{t+1}, p)\right]$$

$$\text{subject to}$$

$$c_t + a_t = a_{t-1}(1+r) + z_t p$$

$$a_t \geq 0$$

$$\ln z_t = \rho \ln z_{t-1} + \epsilon_t$$

- **Q1**: Derive the Euler equation of the model
- **Q2:** Update the EGM algorithm in *household_problem.py* with the new Euler
- **Q3:** Solve the model with 1) $\phi_a = 0$, 2) $\phi_a = 0.1, \sigma = \sigma_a = 1$, 3) $\phi_a = 0.1, \sigma = 1, \sigma_a = 0.7$
  - Compare the normalized policy function c/p
- **Q4:** Conduct an experiment where you permanently redistribute resources across households (change p). What are the aggregate effects across the models?

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